

Simulating Fluid flow in a 2D Lid-Driven Cavity

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(Dated: November 25, 2025)

I. INTRODUCTION

The behavior of fluid flow is inherently complex and often unpredictable, exhibiting intricate patterns and intricate behaviors that can significantly impact various natural and engineered systems. Despite the ubiquity of fluid flow in our daily lives, accurately predicting and understanding its dynamics poses a significant challenge due to the nonlinear and coupled nature of the governing equations that describe fluid motion.

The motion and behavior of fluids, whether liquids or gases, can be described and modeled using the Navier-Stokes equations, a set of coupled nonlinear partial differential equations that govern the conservation of mass, momentum, and energy.

This project aims to solve the Navier-Stokes equation to model fluid flow in a 2D lid-driven cavity. Specifically, the vorticity-stream function formulation of the Navier-Stokes equations is employed to model the intricate flow dynamics within a rectangular enclosure, where one (or more) of the walls (the lid) is set in motion with a constant velocity, while the remaining walls are stationary. What is meant by a vorticity-stream function will be expanded upon in the following sections.

The significance of this research lies in its potential applications across various fields including heat transfer enhancement, mixing processes, and biomechanics. For instance, in heat transfer applications, the presence of obstacles can enhance thermal performance by promoting turbulence and increasing surface area for heat exchange. Similarly, in biomedical engineering, understanding the flow of blood around obstructions or through narrow vessels is essential for developing effective treatments and diagnostic tools.

The computational framework developed in this project can serve as a valuable tool for researchers and engineers to explore and optimize designs involving confined fluid flow with obstacles or obstructions, ultimately leading to improved system performance, energy efficiency, and operational safety.

II. PHYSICS AND NUMERICAL METHODS

The physics and mathematics governing fluids are extensive. So, for the sake of clarity, this section will be divided into subsections that explains all the assumptions and gives a step by step guide towards the final equation. that can be coded to simulate fluid flow

A. Navier-Stokes Equation

The motion and behavior of fluids are governed by the Navier-Stokes equations, which are derived from the fundamental principles of conservation of mass, momentum, and energy. These equations are derived by choosing a infinitesimally small control volume in a liquid, and applying conservation and Newton's Laws to its movement. There are multiple ways to write these equations, and I have chosen to express them with the use of substantial derivatives. Substantial derivative is used to describe the rate of change of a quantity/function in time, if this rate depends on on both spatial and temporal dimensions. For example, for a given control volume of a fluid, the flow depends on how it changes with time. However, if we assume that this control volume is changing positions as well, the flow will now depend on not only the temporal derivative, but the spatial derivative as well. The substantial derivative is denoted with Dq/Dt , if q is the given quantity, and t is used to denote time. So, the set of full, 3D Navier-Stokes equations are[1]:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{V} = 0 \quad (1)$$

$$\frac{\rho D(u)}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + f_x \rho \quad (2)$$

$$\frac{\rho D(v)}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + f_y \rho \quad (3)$$

$$\frac{\rho D(w)}{Dt} = -\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} + f_z \rho \quad (4)$$

$$\begin{aligned} \frac{\rho D \left(e + \frac{1}{2} |\mathbf{V}|^2 \right)}{Dt} &= \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \rho \dot{q} - \frac{\partial(pu)}{\partial x} - \frac{\partial(pv)}{\partial y} - \\ &\quad \frac{\partial(pw)}{\partial z} + \frac{\partial(\tau_{xx}u)}{\partial x} + \frac{\partial(\tau_{yx}u)}{\partial y} + \frac{\partial(\tau_{zx}u)}{\partial z} + \frac{\partial(\tau_{xy}v)}{\partial x} + \frac{\partial(\tau_{yy}v)}{\partial y} + \frac{\partial(\tau_{zy}v)}{\partial z} + \\ &\quad \frac{\partial(\tau_{xz}w)}{\partial x} + \frac{\partial(\tau_{yz}w)}{\partial y} + \frac{\partial(\tau_{zz}w)}{\partial z} + \rho (f_x u + f_y v + f_z w) \end{aligned} \quad (5)$$

Eq.1 is based on conservation of mass. Eq.2, 3 and 4 are based on conservation of x, y, and z directions respectively. Eq.5 is based on energy conservation. u, v, w are the x, y , & z components of the velocity V of the liquid. τ is used to describe the viscous forces acting upon the liquid, and ρ is the density of the liquid. T is the temperature and e is the total energy itself. f is the body forces the liquid may be subjected to.

B. Lid driven Cavity: Assumptions

There are certain assumptions made about the liquid that will help us solve these equations. The assumptions are:

1. It is a 2D domain
2. The liquid is incompressible i.e. no volume, and in turn, density changes.
3. The liquid is viscous
4. No conductive or volumetric heating.
5. No body forces are present

These assumptions make the Navier-Stokes more compact. All the z-dimension terms eliminate, and so do the any temperature changes due to non-viscous changes, body forces, and any derivative of density. So, the Eq.1, 2, 3, 4 and 5 become,

$$\nabla \cdot \mathbf{V} = 0 \quad (6)$$

$$\frac{\rho D(u)}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \quad (7)$$

$$\frac{\rho D(v)}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{vy}}{\partial y} + \frac{\partial \tau_{zv}}{\partial z} \quad (8)$$

$$\begin{aligned} \frac{\rho D \left(e + \frac{1}{2} |\mathbf{V}|^2 \right)}{Dt} &= \frac{\partial(pu)}{\partial x} - \frac{\partial(pv)}{\partial y} + \frac{\partial(\tau_{xx}u)}{\partial x} \\ &+ \frac{\partial(\tau_{vx}u)}{\partial y} + \frac{\partial(\tau_{xy}v)}{\partial x} + \frac{\partial(\tau_{vy}v)}{\partial y} \end{aligned} \quad (9)$$

The equations can be manipulated by using the laws of substantial derivatives. On multiplying Eq.7 by u and Eq.8 by v , and adding them up, we obtain,

$$\frac{\rho D \left(\frac{1}{2} |\mathbf{V}|^2 \right)}{Dt} = -\frac{u \partial(p)}{\partial x} - \frac{v \partial(p)}{\partial y} + \frac{u \partial(\tau_{xx})}{\partial x} + \frac{u \partial(\tau_{yx})}{\partial y} + \frac{v \partial(\tau_{xy})}{\partial x} + \frac{v \partial(\tau_{yy})}{\partial y} \quad (10)$$

where $\mathbf{V}$ is the magnitude of the liquid's velocity of the liquid. At this point, another assumption is made, where it is assumed the liquid is a calorically perfect gas i.e. intermolecular forces do not contribute to the total internal energy, and the specific heat capacity is constant. For an ideal gas, the total energy, e can be expressed as $e = c_v T$. As there is no heating, and the specific heat capacity is constant, the derivative of this would be 0. Hence, due this assumption, Eq.9 would change to look something like

$$\begin{aligned} \frac{\rho D \left(\frac{1}{2} |\mathbf{V}|^2 \right)}{Dt} = & \frac{\partial(pu)}{\partial x} - \frac{\partial(pv)}{\partial y} + \frac{\partial(\tau_{xx}u)}{\partial x} \\ & + \frac{\partial(\tau_{yx}u)}{\partial y} + \frac{\partial(\tau_{xy}v)}{\partial x} + \frac{\partial(\tau_{yy}v)}{\partial y} \end{aligned} \quad (11)$$

As it can be observed, both the equations have identical derivatives on LHS, so these can be equated to each other. After extensive algebra, these equations can be simplified and further split into their respective dimensions, and this results in,

$$\rho_\infty \frac{\partial u}{\partial t} + \rho_\infty \mathbf{V} \cdot \nabla u = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} \quad (12)$$

$$\rho_\infty \frac{\partial v}{\partial t} + \rho_\infty \mathbf{V} \cdot \nabla v = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \quad (13)$$

We switch from ρ to ρ_∞ is done just to denote that the density ρ at a specific point in a liquid is not different from a reference point at infinity, ρ_∞ . This basically means, the density is same everywhere.

C. Variable Change

There shall now be a change of variables introduced to further simplify equations, and express the vector quantities as a single scalar. First off, we begin by using a property of Newtonian fluids, which states that the viscous forces can be expressed as a product of a constant, μ , and the velocity gradient of the fluid itself. Even though a mathematical proof for this property can be worked out, we will accept this as a fundamental fact for the sake

of this report. These viscous force work out to be,

$$\tau_{xx} = \frac{2\mu}{3} \left[2 \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right] \quad (14)$$

$$\tau_{xy}, \tau_{yx} = \mu \left[\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right] \quad (15)$$

$$\tau_{yy} = \frac{2\mu}{3} \left[2 \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right] \quad (16)$$

Also, we know that, according to Eq.6, that $\nabla \cdot \dot{V} = 0$ which, when expanded out, says

$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} &= 0 \\ \frac{\partial u}{\partial x} &= -\frac{\partial v}{\partial y} \end{aligned} \quad (17)$$

Now, we substitute Eq. 17 into Eq.13, and then substitute the values back into Eq. 12 and 13. These steps are, again, algebraically extensive, so the complete derivation shall be skipped, and the final equations can be presented to be

$$\rho_{\infty} \frac{\partial u}{\partial t} + \rho_{\infty} u \frac{\partial u}{\partial x} + \rho_{\infty} v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (18)$$

$$\rho_{\infty} \frac{\partial v}{\partial t} + \rho_{\infty} u \frac{\partial v}{\partial x} + \rho_{\infty} v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (19)$$

D. Stream-function Vorticity Formulation

The curl for the flow velocity can be given by, $\nabla \times V = \omega$, where ω is the vorticity. When expanded, this gives us,

$$\omega = \left[\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right] \vec{k} \quad (20)$$

Now, for our last, and probably the most important step for this simplification, we shall introduce the stream-function, ψ . Physically, ψ can be defined as the quantity of fluid moving across some convenient imaginary line in the flow pattern, and lines of constant stream function (amount of flow or flux) may be plotted to give a picture of the flow pattern. ψ is scalar quantity, and this stream-function chosen such that,

$$u = \frac{\partial \psi}{\partial y} \quad (21)$$

$$v = -\frac{\partial \psi}{\partial x} \quad (22)$$

We can use these equations reformulate Eq.20,

$$-\omega = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \quad (23)$$

A few mathematical manipulations are now applied here. We can take the derivative of Eq. 18 with respect to x, and derivative Eq.19 with respect to y. We will also use Eq.6 , i.e. divergence of velocity is 0, to get, .

$$\rho_\infty \frac{\partial \omega}{\partial t} + \rho_\infty \frac{\partial \psi}{\partial y} \frac{\partial \omega}{\partial x} + \rho_\infty \frac{\partial \psi}{\partial x} \frac{\partial \omega}{\partial y} = \mu \frac{\partial^2 \omega}{\partial x^2} + \mu \frac{\partial^2 \omega}{\partial y^2} \quad (24)$$

To obtain the final set of equations, we non-dimensionalize Eq.24, by dividing through by the density ρ_∞ , a reference velocity and a reference length. On doing so, we obtain the final set of equations that we can apply boundary conditions to and solve using Python. which are,

$$\frac{\partial \omega}{\partial t} + \frac{\partial \psi}{\partial y} \frac{\partial \omega}{\partial x} + \frac{\partial \psi}{\partial x} \frac{\partial \omega}{\partial y} = \frac{\mu}{\rho_\infty U_{ref} L} \left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right) \quad (25)$$

$$-\omega = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \quad (26)$$

As a consequence this non-dimensionalization, we obtain $\frac{\mu}{\rho_\infty U_{ref} L}$ on the right side of the equation, which is the inverse of the Reynold's number, which can be approximated for a given fluid and the dimensions of the cavity it is entrapped in. We can now introduce our problem and boundary conditions to code in Python.

E. Problem and Boundary Conditions

Before introducing the problem, I would like the reader to note that after the entire process is worked out, any changes in the boundary condition would not require large changes in the code itself, rather certain minor tweaks can be made to introduce these changes. The problem serves as a starting point for coding the problem up. Here is our problem: a fluid is entrapped in an impenetrable cavity of 2×2 meters, the upper boundary (the lid) at $y = 1$, moves with a constant velocity $U = 1$. The Reynolds number based on the size of the domain, the velocity of the moving wall, density $\rho = 1$ and viscosity $\mu = 0.005$ is $Re = 200$. Fig.1 serves as a visual reference. The boundary conditions given are in terms of u and v, so they need to be converted to ψ and ω . On the left edge, as the height y changes, v remains

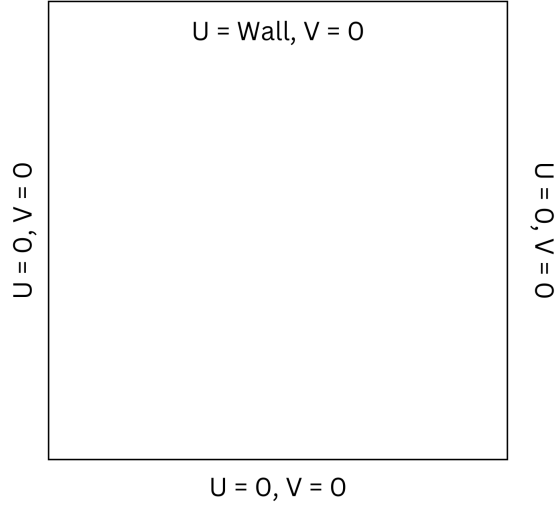


FIG. 1: Boundary Conditions for a 2D lid-driven cavity.

zero. Hence $\frac{\partial \psi}{\partial y} = 0$. So, ψ must be a constant. Due to continuity across the boundaries (in a physical sense, we know the corners are shared by two edges, hence continuity must hold) ψ must be equal. As ψ is a constant, it can assume any arbitrary initial value, which we shall assume to be 0. Applying the boundary conditions to ω , we know $-\omega = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}$. On the left wall at some height j , $\frac{\partial^2 \psi}{\partial y^2} = 0$. So the boundary condition looks like,

$$\omega_{1,j} = - \left. \frac{\partial^2 \psi}{\partial x^2} \right|_{1,j} \quad (27)$$

To obtain a finite difference approximation for the second-order derivative in the equation above, consider the Taylor series expansion

$$\psi_{2,j} = \psi_{1,j} + \left. \frac{\partial \psi}{\partial x} \right|_{1,j} \Delta x + \left. \frac{\partial^2 \psi}{\partial x^2} \right|_{1,j} \frac{\Delta x^2}{2} + \dots \quad (28)$$

Along left wall,

$$\left. \frac{\partial \psi}{\partial x} \right|_{1,j} = -v_{1,j} \quad (29)$$

Therefore, Taylor series expansion can be rearranged to get,

$$\left. \frac{\partial^2 \psi}{\partial x^2} \right|_{1,j} = \frac{2(\psi_{2,j} - \psi_{1,j})}{\Delta x^2} + \frac{2v_{1,j}}{\Delta x} \quad (30)$$

Substitution of Eq.30 into 29 yields,

$$\omega_{1,j} = \frac{2(\psi_{1,j} - \psi_{2,j})}{\Delta x^2} - \frac{2v_{1,j}}{\Delta x} \quad (31)$$

A similar procedure is used to derive the boundary conditions at right, bottom, and top wall. The appropriate expressions are,

$$\begin{aligned}
\omega_{M,j} &= - \left. \frac{\partial^2 \psi}{\partial x^2} \right|_{M,j} = \frac{2(\psi_{M,j} - \psi_{M-1,j})}{\Delta x^2} + \frac{2v_{M,j}}{\Delta x} \\
\omega_{i,1} &= - \left. \frac{\partial^2 \psi}{\partial y^2} \right|_{i,1} = \frac{2(\psi_{i,1} - \psi_{i,2})}{\Delta y^2} + \frac{2u_{i,1}}{\Delta y} \\
\omega_{i,N} &= - \left. \frac{\partial^2 \psi}{\partial y^2} \right|_{i,N} = \frac{2(\psi_{i,N} - \psi_{i,N-1})}{\Delta y^2} - \frac{2u_{i,N}}{\Delta y}
\end{aligned} \tag{32}$$

where M is the maximum length of the grid, N is the maximum height of the grid, and the subscripts i, j denote the grid point being worked with. The superscript n denotes that the given quantity is in present time step. The reason for expressing the ω boundary condition in this way was so that the boundary conditions be coded in as a single-order partial derivative.

F. Code

The code uses the finite difference method learnt in class to solve the partial differential equations. To start off we set up the the cavity, and decide the the number of steps and time steps. Then, inside a for loop that runs for the number of steps defined, we define our boundary conditions for ω (Point to remember: ψ is 0 at all 4 boundaries).

Now, it can be observed in Eq.25 that only one quantity needs to be updated with time which is ω . So the code is used to solve for $\partial\omega/\partial t$. The amazing part of the code is the fact that every other quantity is known so we can solve for ω at time step $n+1$, while all the other quantities are at time step n , and are known. Then using this ω at $n+1$, we update the psi values. Here, the jacobi method is used to calculate the differencing scheme for the double derivative.[2] Again, the derivation of this shall not be explained, as it is out of the scope for this project, but this method was a recommendation in multiple books to be used for any finite differencing scheme, whether being solved computationally or not.

On applying the finite difference to Eq.25, it results in,

$$\begin{aligned}
\frac{\omega_{i,j}^{n+1} - \omega_{i,j}^n}{\Delta t} &= \frac{1}{Re} \left[\frac{\omega_{i+1,j} - 2\omega_{i,j} + \omega_{i-1,j}}{(\Delta x)^2} + \frac{\omega_{i,j+1} - 2\omega_{i,j} + \omega_{i,j-1}}{(\Delta y)^2} \right] - \\
&\quad \left(\frac{\psi_{i,j+1} - \psi_{i,j-1}}{2\Delta y} \right) \left(\frac{\omega_{i+1,j} - \omega_{i-1,j}}{2\Delta x} \right) \\
&\quad - \left(\frac{\psi_{i+1,j} - \psi_{i-1,j}}{2\Delta x} \right) \left(\frac{\omega_{i,j+1} - \omega_{i,j-1}}{2\Delta y} \right)
\end{aligned} \tag{33}$$

On applying the finite difference method to the Eq.26, it results in.

$$\psi_{i,j}^{n+1} \left[\frac{-2}{(\Delta x)^2} - \frac{2}{(\Delta y)^2} \right] = -\omega_{i,j}^{n+1} - \left[\frac{\psi_{i+1,j}^n + \psi_{i-1,j}^n}{(\Delta x)^2} \right] - \left[\frac{\psi_{i,j+1}^n + \psi_{i,j-1}^n}{(\Delta y)^2} \right] \quad (34)$$

These equations are what are finally coded into our program. At the end of every iteration, velocities were also calculated using the derivative of ψ with respect to x and y respectively.

G. Von-Neumann Stability

The stability of any finite difference scheme depends on the time interval used. Usually this can be calculated through extensive algebra, but in this case trial and error was used to determine the ideal time stepping, by using multiple values and determining at what time interval the code worked well while also being small enough to provide a good amount of information over varying periods of time.

H. Obstacle in the Cavity

As mentioned earlier, the introduction of big changes would require only a few small tweaks in the code itself. Here we decided to introduce, 1) a square box of size 0.2×0.2 in the center of the cavity and 2) a circular obstacle of radius 0.2 in the center of the vortex. The units have been dropped, as the equation was non-dimensionalized.

To introduce these obstacles, first a mask was created in the grid itself based on the area and position of the obstacle. Then, in the for loop, the value for ω and ψ was updated to be 0 at the end of every iteration. What this essentially did was work out ω and ψ while abiding to the condition that the flow inside the obstacle remains 0.

III. VERIFICATION

The end result was plotted using streamplot, that plotted the velocities, u and v as field lines over 2D space. Multiple methods were used to verify the code. First, the number of steps was decreased to 100, so that the total time that had elapsed after the simulation had begun was 0.05 seconds. In this case, we should observe little to no movement in the fluid. This is what was observed in the plot, as can be seen in Fig.2.

Then, the number of steps were changed to simulate the situation for a time period of

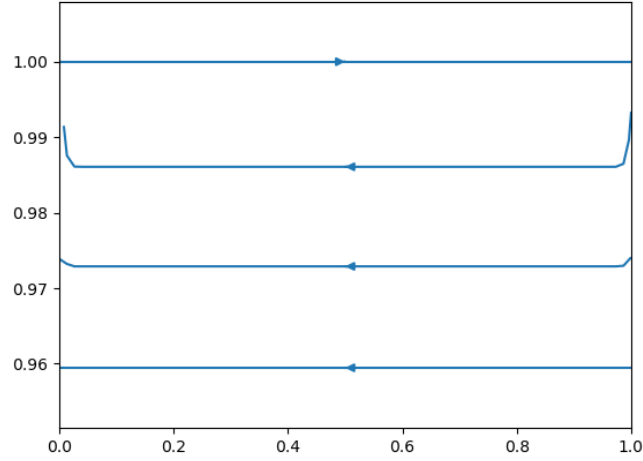


FIG. 2: Plot for velocity field lines after 0.05 seconds. There is little movement in the fluid.

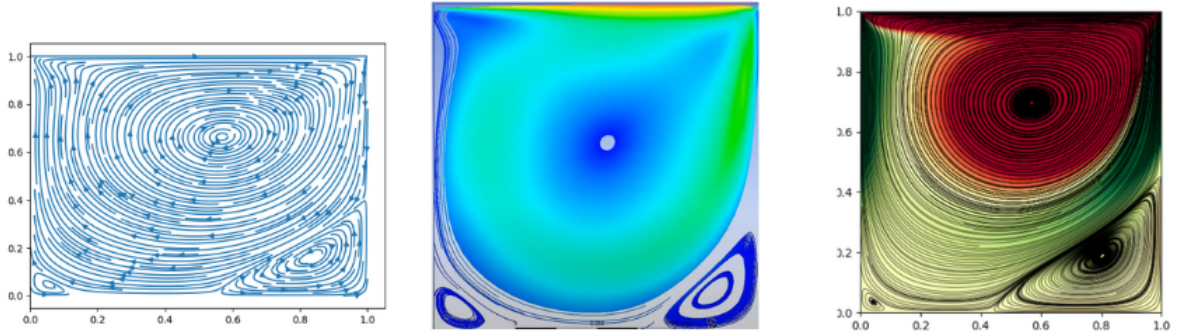


FIG. 3: Left: my simulation, Center and Right: Simulations from ANSYS and MATLAB CFD solvers. As we can see, similar vortex formation is seen in similar locations

[3]

10 seconds. As this is a very fundamental part of fluid dynamics, multiple simulations exist for such a scenario. When compared to these simulations, our resultant plot looks identical. There were some differences in the cavity defined and the boundary conditions for these simulations, but overall, same phenomenon of a triple vortex formation is observed in similar locations which shall be discussed more in the results section. See Fig.3 for comparison. There were no simulations that modeled fluid flow in a 2D lid-driven cavity for when an obstacle was kept in cavity, so those plots could not be cross-verified.

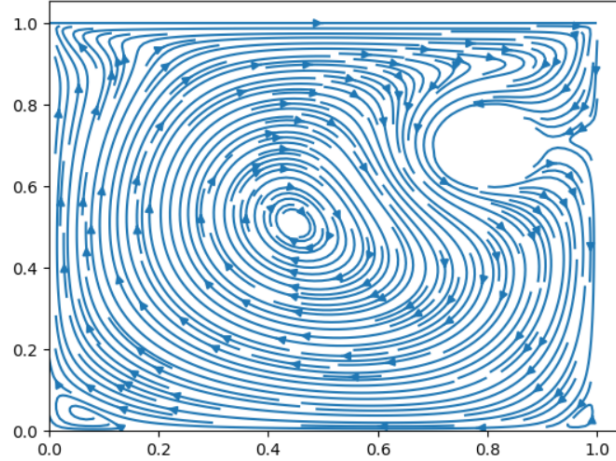


FIG. 4: A circular obstacle kept at the center of the vortex, and as we can see the center is shifted, eliminating the secondary vortices as well.

So, to verify those plots, I changed the position and the shape of the second obstacle to be a circle in the center of where the vortex was. In this case, the center of the vortex should ideally shift, which it did, as can be seen in Fig.4. It also eliminated the secondary vortices which would be an intuitive result.

IV. RESULTS AND ANALYSIS

A. Lid-Driven capacity with no Obstacle

The numerical simulations of two-dimensional incompressible viscous fluid flow in a lid-driven cavity with obstacles, conducted using the vorticity-stream function formulation of the Navier-Stokes equations, have yielded insightful results that provide a comprehensive understanding of the intricate flow dynamics within the domain.

to see. Fig.5 depicts the streamline patterns and vorticity distribution for the lid-driven cavity flow without any obstacles at a Reynolds number of 200. The formation of a primary vortex in the central-right of the cavity and two secondary counter-rotating vortices in the bottom corners is clearly visible. The formation of counter-rotating vortices is unintuitive, hence surprising Fig. 6 shows a time evolution of vortex formation after $t=1s, 5s, 7s, 10s$.

Fig. 7 illustrates the effect of introducing a square obstacle of side length 0.2 units, positioned at the center of the cavity ($x = 0.4, y = 0.4$). The presence of the obstacle

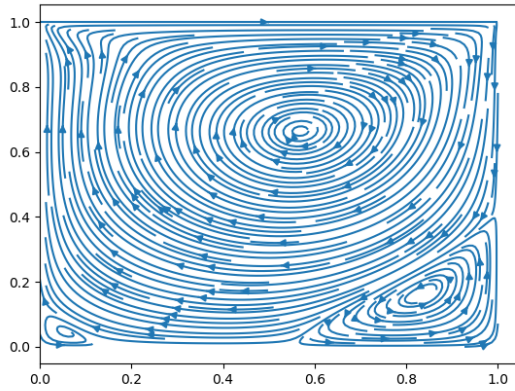


FIG. 5: Vorticity distribution of a liquid of Reynolds Number 200, after 10 seconds

significantly alters the flow dynamics, resulting in the formation of additional vortices and recirculation zones around the obstacle. The streamlines exhibit intricate patterns, with fluid particles flowing around the obstacle and creating complex vortex structures in its wake.

The influence of obstacle position on the flow dynamics was further investigated by simulating the lid-driven cavity flow with a circular obstacle of radius 0.1 units placed in the top right corner of the domain ($x = 0.8$, $y = 0.7$). Fig. 8 illustrates the streamline patterns and vorticity distribution for this configuration. Notably, the presence of the obstacle in the top right corner induces a strong recirculation zone in the center of the cavity, and the elimination of most of the secondary vortices.

V. CONCLUSION

The present study has successfully developed and implemented a numerical model based on the vorticity-stream function formulation of the Navier-Stokes equations to simulate two-dimensional incompressible viscous fluid flow in a lid-driven cavity with obstacles. The results obtained from this investigation provide valuable insights into the intricate flow dynamics and vortex structures that arise in the presence of obstacles within the flow domain. The simulations conducted for the lid-driven cavity flow without obstacles serve as a benchmark, exhibiting the characteristic formation of a primary vortex in the center and two secondary counter-rotating vortices in the bottom corners, as expected for this canonical

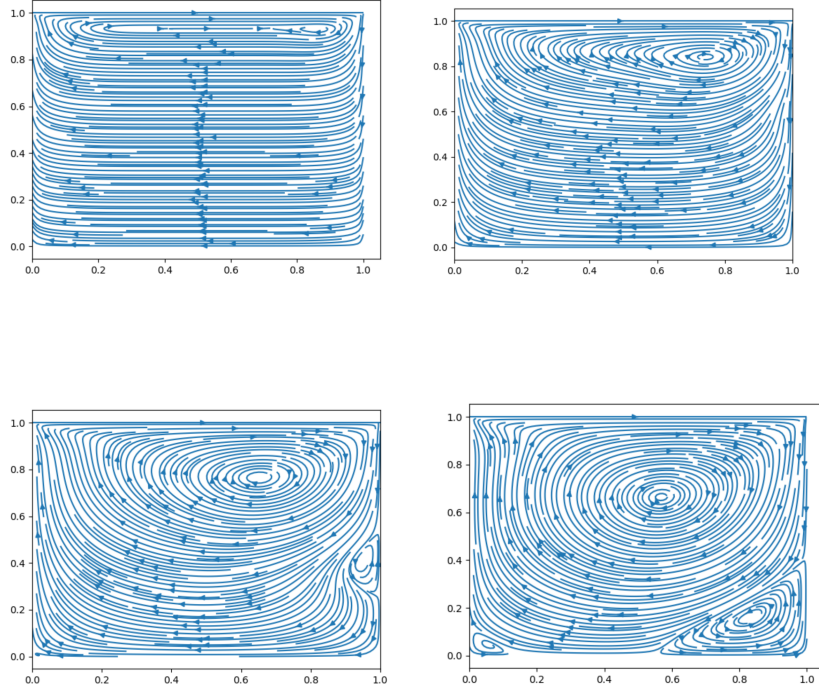


FIG. 6: Top Left: 1s — Top right: 5s — Bottom Left: 7s — Bottom right: 10s

problem. The introduction of obstacles, both square and circular, significantly alters the flow patterns, leading to the generation of additional vortices, recirculation zones, and intricate streamline patterns.

While this study focused on two-dimensional simulations of incompressible viscous fluid flow, future work could extend the numerical model to three-dimensional configurations, incorporate turbulence models, or investigate the effects of time-dependent boundary conditions or obstacle motion. Additionally, experimental validation of the numerical results would further strengthen the credibility of the computational framework and provide insights into potential improvements or refinements.

[1] *Navier-Stokes Equations* — [grc.nasa.gov](https://www.grc.nasa.gov/www/k-12/airplane/nseqs.html), <https://www.grc.nasa.gov/www/k-12/airplane/nseqs.html>, [Accessed 02-05-2024].

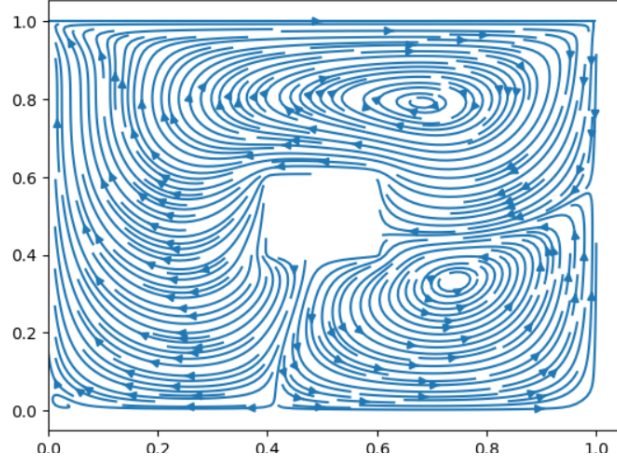


FIG. 7: Square obstacle in the center

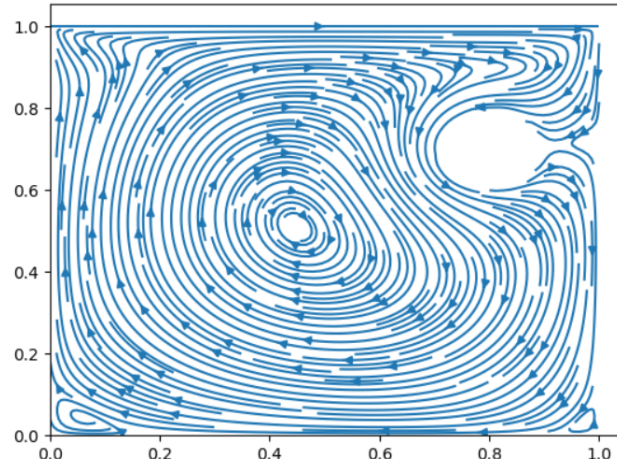


FIG. 8: A circular obstacle in the top right of the vortex

- [2] H. Moghaderi, M. Dehghan, and M. Hajarian, *Numerical Algorithms* **72**, 483 (2016).
- [3] *ANSYS CFD Tutorial: Lid Driven Cavitation* — *youtube.com*, <https://www.youtube.com/watch?v=L2W6pcHvjLQ>, [Accessed 02-05-2024].